

A Standalone Companion To Fixed Sparse-Grid Quadrature Filtering And Its Analytical Gradient

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Abstract

This note develops fixed sparse-grid quadrature filtering as a standalone high-dimensional filtering proposal. The construction starts from the sparse-grid quadrature nonlinear filter of Jia, Xin, and Cheng [1], expands the Gaussian projection and sparse-grid formulas in BayesFilter notation, and then defines a fixed approximate likelihood scalar with an analytical same-scalar gradient. The scalar is a deterministic Gaussian-projection likelihood: it is a candidate fixed target for HMC experiments when branch diagnostics remain stable, but it is not the exact nonlinear filtering likelihood and it does not store the full posterior density. The final section compares this proposal with the Zhao–Cui squared tensor-train proposal [3]: fixed SGQF is transparent and fixed-rule differentiable, while squared TT is a richer non-Gaussian density approximation.

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1 What This Note Computes

The object computed here is deliberately narrow. At time t , before observing y_t , the filter carries one Gaussian approximation

$$X_{t-1} \mid y_{1:t-1} \approx \mathcal{N}(m_{t-1}, P_{t-1}). \quad (1.1)$$

It then uses a deterministic sparse-grid quadrature rule to approximate the Gaussian moments needed by a Kalman-style projection update. The output after observing y_t is again only

$$X_t \mid y_{1:t} \approx \mathcal{N}(m_t, P_t), \quad (1.2)$$

together with the approximate Gaussian innovation log likelihood

$$\hat{\ell}_t = -\frac{1}{2} \left\{ \log \det S_t + v_t^\top S_t^{-1} v_t + n_y \log(2\pi) \right\}. \quad (1.3)$$

The fixed-SGQF proposal is to use the sum

$$\hat{\ell}_T^{\text{FSGQ}}(\theta) = \sum_{t=1}^T \hat{\ell}_t^{\text{FSGQ}}(\theta) \quad (1.4)$$

as a deterministic approximate likelihood scalar and to differentiate that same scalar analytically.

This note therefore computes an approximate likelihood and a Gaussian projection recursion. It does not compute the full nonlinear posterior law. Sparse-grid exactness is exactness of a quadrature formula for selected Gaussian-weighted polynomial integrands. It is not a guarantee that the posterior is Gaussian, unimodal, or accurately represented by (m_t, P_t) .

A scalar cell that keeps us honest. Let

$$X \sim \mathcal{N}(0, P), \quad Y = X^2 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R). \quad (1.5)$$

For $y > 0$ and small R , the exact posterior density is proportional to

$$\exp \left[-\frac{x^2}{2P} - \frac{(y - x^2)^2}{2R} \right], \quad (1.6)$$

which can have two separated lobes near $+\sqrt{y}$ and $-\sqrt{y}$. A Gaussian projection filter can compute accurate moments of X^2 , but after the update it still stores only one Gaussian. This is not a bug in the quadrature rule. It is the modeling choice in (1.2).

The same limitation explains the role of fixed SGQF in the high-dimensional program. It is a transparent fixed approximate likelihood target with a well-defined gradient. It is not the non-Gaussian density approximation lane; that role belongs to squared tensor-train filtering.

2 State-Space Model And Exact Filtering Recursion

Use the additive-noise state-space model

$$X_t = f_\theta(X_{t-1}) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_\theta), \quad (2.1)$$

$$Y_t = h_\theta(X_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_\theta), \quad (2.2)$$

with $X_t \in \mathbb{R}^{n_x}$, $Y_t \in \mathbb{R}^{n_y}$, and parameter $\theta \in \mathbb{R}^p$. The exact Bayesian prediction and update are

$$p_\theta(x_t | y_{1:t-1}) = \int p_\theta(x_t | x_{t-1}) p_\theta(x_{t-1} | y_{1:t-1}) dx_{t-1}, \quad (2.3)$$

$$p_\theta(x_t | y_{1:t}) = \frac{g_\theta(y_t | x_t) p_\theta(x_t | y_{1:t-1})}{\int g_\theta(y_t | x_t) p_\theta(x_t | y_{1:t-1}) dx_t}. \quad (2.4)$$

These are the conceptual filtering equations used by Gaussian approximation filters in Jia, Xin, and Cheng [1, Section 2]. In nonlinear models, the integrals in (2.3) and (2.4) usually do not close in finite-dimensional form. Fixed SGQF replaces those exact densities by a Gaussian moment projection.

3 Gaussian Projection From Moments

Assume the predictive state is represented by

$$X_t^- \sim \mathcal{N}(m_t^-, P_t^-). \quad (3.1)$$

Let

$$Z_t = h_\theta(X_t^-) \quad (3.2)$$

be the noiseless predicted observation. The Gaussian projection needs

$$\bar{z}_t = \mathbb{E}[Z_t], \quad (3.3)$$

$$S_t = R_\theta + \mathbb{E}[(Z_t - \bar{z}_t)(Z_t - \bar{z}_t)^\top], \quad (3.4)$$

$$C_{xz,t} = \mathbb{E}[(X_t^- - m_t^-)(Z_t - \bar{z}_t)^\top]. \quad (3.5)$$

Given these moments, define

$$v_t = y_t - \bar{z}_t, \quad K_t = C_{xz,t} S_t^{-1}, \quad (3.6)$$

$$m_t = m_t^- + K_t v_t, \quad P_t = P_t^- - K_t S_t K_t^\top. \quad (3.7)$$

Proposition 1 (Affine projection). *Among estimators of X_t^- of the form $a + BY_t$, the minimum mean-square estimator under the moments in (3.3)–(3.5) uses $B = C_{xz,t} S_t^{-1}$. Storing the resulting mean and covariance as a Gaussian gives (3.7).*

Proof. Center $\tilde{X} = X_t^- - m_t^-$ and $\tilde{Y} = Y_t - \bar{z}_t$. The squared-error objective for $m_t^- + B\tilde{Y}$ is

$$J(B) = \mathbb{E}\|\tilde{X} - B\tilde{Y}\|^2 = \text{tr}(P_t^-) - 2\text{tr}(BC_{xz,t}^\top) + \text{tr}(BS_tB^\top). \quad (3.8)$$

Differentiating gives $\nabla_B J = -2C_{xz,t} + 2BS_t$. If S_t is nonsingular, the stationary point is $B = C_{xz,t}S_t^{-1}$. Substitution gives the residual covariance $P_t^- - C_{xz,t}S_t^{-1}C_{xz,t}^\top$, which equals (3.7). $\square$

The approximation has two layers. First, the moments (3.3)–(3.5) may be approximated by quadrature. Second, the resulting posterior information is compressed to one Gaussian.

4 One-Dimensional Gaussian Quadrature And Tensor Products

Let $\xi \sim \mathcal{N}(0, 1)$. A one-dimensional standard-normal quadrature rule at level ℓ is a set of nodes and weights

$$I_\ell[\psi] = \sum_{a=1}^{s_\ell} \omega_{\ell,a} \psi(\zeta_{\ell,a}) \approx \int_{\mathbb{R}} \psi(\xi) \phi(\xi) d\xi. \quad (4.1)$$

For a multi-index $\mathbf{i} = (i_1, \dots, i_b)$, the corresponding tensor rule is

$$I_{\mathbf{i}}[F] = (I_{i_1} \otimes \dots \otimes I_{i_b})[F]. \quad (4.2)$$

Its nodes and weights are

$$\xi^{\mathbf{i},\alpha} = (\zeta_{i_1,\alpha_1}, \dots, \zeta_{i_b,\alpha_b}), \quad (4.3)$$

$$w^{\mathbf{i},\alpha} = \prod_{j=1}^b \omega_{i_j,\alpha_j}, \quad \alpha_j \in \{1, \dots, s_{i_j}\}. \quad (4.4)$$

A full tensor-product rule is simple but expensive: if every coordinate uses s nodes, then $M = s^b$. Jia, Xin, and Cheng introduce the sparse-grid combination to reduce this point count for fixed accuracy level [1, Section 3].

5 Sparse-Grid Rule Reconstructed

For dimension b and level L , define the level band

$$\mathcal{A}_{b,L} = \{\mathbf{i} \in \mathbb{N}^b : L \leq |\mathbf{i}| \leq L + b - 1\}, \quad |\mathbf{i}| = \sum_{j=1}^b i_j. \quad (5.1)$$

For each $\mathbf{i} \in \mathcal{A}_{b,L}$, set

$$c_{\mathbf{i}} = (-1)^{L+b-1-|\mathbf{i}|} \binom{b-1}{L+b-1-|\mathbf{i}|}. \quad (5.2)$$

The sparse-grid quadrature approximation is

$$A_{b,L}[F] = \sum_{\mathbf{i} \in \mathcal{A}_{b,L}} c_{\mathbf{i}} I_{\mathbf{i}}[F]. \quad (5.3)$$

This is the BayesFilter notation for the Jia–Xin–Cheng sparse-grid formula [1, Eq. 26–29]. Their notation groups indices by $|\mathbf{i}| = b + q$; the band $q = L - b, \dots, L - 1$ is equivalent to (5.1).

From formula to cloud. Equation (5.3) is not yet an implementation object. The same node can appear in several tensor rules. The fixed cloud is the accumulated dictionary

$$D[\xi] = \sum_{\mathbf{i}, \alpha: \xi^{\mathbf{i}, \alpha} = \xi} c_{\mathbf{i}} w^{\mathbf{i}, \alpha}. \quad (5.4)$$

After merging duplicate standardized nodes, enumerate the nonzero entries as

$$\mathcal{C}_{b,L} = \{(\xi^{(r)}, w_r)\}_{r=1}^{M_{b,L}}. \quad (5.5)$$

Jia, Xin, and Cheng describe this point-and-weight generation in Algorithm 1 [1, Algorithm 1]. For the fixed BayesFilter scalar, the cloud $\mathcal{C}_{b,L}$, the one-dimensional rules, the merge tolerance, and the accumulated signed weights are part of the mathematical target.

Source exactness and point count. The source exactness statement is a quadrature statement: under the univariate conditions in Jia et al. [1, Theorem 3.1], the sparse-grid rule inherits a polynomial exactness level. This does not imply exact nonlinear posterior filtering. The point-count statements in Jia et al. [1, Proposition 3.1–3.2] are fixed-level statements. They support the claim that sparse grids can grow polynomially in dimension for fixed level, not the claim that a given fixed level is accurate for every high-dimensional nonlinear posterior.

UKF relation. Jia, Xin, and Cheng prove that an unscented-transform point set is a member of the SGQF family at level 2 under their parameterization [1, Theorem 3.2]. In this note, the relation is used only as orientation: fixed SGQF generalizes the idea of deterministic Gaussian moment points, but the selected BayesFilter target is the declared fixed cloud (5.5).

6 A Toy Fixed Grid With Duplicate Merging

This section makes the sparse-grid cloud concrete. Take $b = 2$ and $L = 2$. Use

$$I_1 : (0, 1), \quad I_2 : \left\{(-\sqrt{3}, 1/6), (0, 2/3), (\sqrt{3}, 1/6)\right\}, \quad (6.1)$$

which is the familiar three-point standard-normal rule at level 2. The band $\mathcal{A}_{2,2}$ contains

$$(1, 1), \quad (1, 2), \quad (2, 1), \quad (6.2)$$

with coefficients

$$c_{(1,1)} = -1, \quad c_{(1,2)} = 1, \quad c_{(2,1)} = 1. \quad (6.3)$$

Before merging, the node $(0, 0)$ appears three times. Its accumulated weight is

$$-1 + \frac{2}{3} + \frac{2}{3} = \frac{1}{3}. \quad (6.4)$$

The four axis nodes each receive weight $1/6$. The merged cloud is

node ξ	accumulated weight	source components
$(0, 0)$	$1/3$	$(1, 1), (1, 2), (2, 1)$
$(-\sqrt{3}, 0)$	$1/6$	$(2, 1)$
$(\sqrt{3}, 0)$	$1/6$	$(2, 1)$
$(0, -\sqrt{3})$	$1/6$	$(1, 2)$
$(0, \sqrt{3})$	$1/6$	$(1, 2)$

The weights sum to one:

$$\frac{1}{3} + 4 \cdot \frac{1}{6} = 1. \quad (6.5)$$

This toy cloud illustrates two implementation rules that are easy to get wrong. First, duplicate nodes must be accumulated before the cloud is used. Second, signed sparse-grid coefficients are applied before accumulation; a future level or a different univariate family can produce negative accumulated weights, so covariance matrices require explicit diagnostics.

7 FixedSGQF Filtering Value Path

Fix a standardized sparse-grid cloud

$$\mathcal{C} = \{(\xi^{(r)}, w_r)\}_{r=1}^M, \quad \xi^{(r)} \in \mathbb{R}^{n_x}, \quad w_r \in \mathbb{R}. \quad (7.1)$$

The cloud is frozen before likelihood evaluation. The value path has two quadrature stages at each time.

7.1 Prediction Stage

Let $P_{t-1} = C_{t-1}C_{t-1}^\top$ on the declared square-root branch. Place the previous Gaussian cloud:

$$\chi_{t-1}^{(r)} = m_{t-1} + C_{t-1}\xi^{(r)}. \quad (7.2)$$

Push points through the transition:

$$a_t^{(r)} = f_\theta(\chi_{t-1}^{(r)}). \quad (7.3)$$

The predicted Gaussian is

$$m_t^- = \sum_{r=1}^M w_r a_t^{(r)}, \quad (7.4)$$

$$P_t^- = Q_\theta + \sum_{r=1}^M w_r (a_t^{(r)} - m_t^-)(a_t^{(r)} - m_t^-)^\top. \quad (7.5)$$

After (7.5), the implementation replaces P_t^- by $(P_t^- + P_t^{-\top})/2$ and checks the declared positive-definite branch. For the differentiable lane in this note, there is no adaptive jitter, floor, clipping, pivot, or eigenvalue repair inside the scalar. If the check fails, the value path returns a veto rather than quietly changing the scalar.

7.2 Observation Stage

Factor $P_t^- = C_t^-(C_t^-)^\top$ on the declared branch and place the predictive cloud:

$$\chi_t^{(r)} = m_t^- + C_t^-\xi^{(r)}. \quad (7.6)$$

Evaluate the observation map:

$$z_t^{(r)} = h_\theta(\chi_t^{(r)}). \quad (7.7)$$

Then compute

$$\bar{z}_t = \sum_{r=1}^M w_r z_t^{(r)}, \quad (7.8)$$

$$\Delta_t^{(r)} = z_t^{(r)} - \bar{z}_t, \quad (7.9)$$

$$S_t = R_\theta + \sum_{r=1}^M w_r \Delta_t^{(r)} \Delta_t^{(r)\top}, \quad (7.10)$$

$$C_{xz,t} = \sum_{r=1}^M w_r (\chi_t^{(r)} - m_t^-) \Delta_t^{(r)\top}. \quad (7.11)$$

The innovation and log likelihood are

$$v_t = y_t - \bar{z}_t, \quad (7.12)$$

$$\widehat{\ell}_t^{\text{FSGQ}} = -\frac{1}{2} \left\{ \log \det S_t + v_t^\top S_t^{-1} v_t + n_y \log(2\pi) \right\}. \quad (7.13)$$

If S_t is not positive definite on the declared branch, the value path returns a veto. If it passes, the Gaussian projection update is

$$K_t = C_{xz,t} S_t^{-1}, \quad (7.14)$$

$$m_t = m_t^- + K_t v_t, \quad (7.15)$$

$$P_t = P_t^- - K_t S_t K_t^\top. \quad (7.16)$$

8 The Saved Scalar And Same-Scalar Contract

The derivative in this note is meaningful only for the scalar actually evaluated. A fixed-SGQF branch is the tuple

$$\begin{aligned} \mathcal{B} = & (y_{1:T}, \text{ observation preprocessing, } m_0(\theta), P_0(\theta), \dot{m}_0, \dot{P}_0, \\ & \mathcal{C}, \text{ one-dimensional rules, duplicate-merge tolerance, } C(\cdot), \\ & \text{symmetrize-then-veto rule, } f_\theta, h_\theta, Q_\theta, R_\theta). \end{aligned} \quad (8.1)$$

The fixed scalar is

$$\widehat{\ell}_T^{\text{FSGQ}}(\theta; \mathcal{B}) = \sum_{t=1}^T \widehat{\ell}_t^{\text{FSGQ}}(\theta; \mathcal{B}), \quad (8.2)$$

where every term is computed by (7.2)–(7.16). The same $\mathcal{B}$ must be used for values, gradients, and finite differences. Changing the grid level, active index set, duplicate-merge tolerance, Cholesky pivot, observation preprocessing, initial-condition policy, eigenvalue floor, covariance clipping, or point pruning changes the scalar. The derivative formulas below cover the open branch where the symmetrized covariances pass the Cholesky and positive-definite checks without repair. If a jitter, floor, clipping, or pivoting map is desired, it must be declared as a different scalar with its own derivative or stop rule.

9 Analytical Gradient Of The Fixed Scalar

Differentiate with respect to parameter component θ_i . A dot denotes ∂_i , for example $\dot{m}_t = \partial_i m_t$. The data y_t are recorded and do not depend on θ .

9.1 Square-Root Branch

For the default unpivoted Cholesky branch, $P = LL^\top$ with positive diagonal and $C(P) = L$. Given symmetric $\dot{P}$, set

$$A = L^{-1}\dot{P}L^{-\top}. \quad (9.1)$$

Define the lower-triangular operator

$$\Phi(A)_{jk} = \begin{cases} A_{jk}, & j > k, \\ A_{jj}/2, & j = k, \\ 0, & j < k. \end{cases} \quad (9.2)$$

Then

$$\dot{L} = L\Phi(A). \quad (9.3)$$

Indeed, $\Phi(A) + \Phi(A)^\top = A$, so

$$\dot{L}L^\top + L\dot{L}^\top = L\{\Phi(A) + \Phi(A)^\top\}L^\top = \dot{P}. \quad (9.4)$$

If a different square-root map is used, the implementation must provide $\dot{C} = DC(P)[\dot{P}]$ for the same branch.

9.2 Prediction Sensitivities

From (7.2),

$$\dot{\chi}_{t-1}^{(r)} = \dot{m}_{t-1} + \dot{C}_{t-1}\xi^{(r)}. \quad (9.5)$$

From (7.3),

$$\dot{a}_t^{(r)} = D_x f_\theta(\chi_{t-1}^{(r)})\dot{\chi}_{t-1}^{(r)} + \partial_i f_\theta(\chi_{t-1}^{(r)}). \quad (9.6)$$

The predicted mean derivative is

$$\dot{m}_t^- = \sum_{r=1}^M w_r \dot{a}_t^{(r)}. \quad (9.7)$$

Let

$$d_t^{(r)} = a_t^{(r)} - m_t^-, \quad \dot{d}_t^{(r)} = \dot{a}_t^{(r)} - \dot{m}_t^-. \quad (9.8)$$

Then

$$\dot{P}_t^- = \dot{Q}_\theta + \sum_{r=1}^M w_r \left\{ \dot{d}_t^{(r)} d_t^{(r)\top} + d_t^{(r)} \dot{d}_t^{(r)\top} \right\}. \quad (9.9)$$

9.3 Observation Sensitivities

Differentiate (7.6):

$$\dot{\chi}_t^{(r)} = \dot{m}_t^- + \dot{C}_t^- \xi^{(r)}. \quad (9.10)$$

Then

$$\dot{z}_t^{(r)} = D_x h_\theta(\chi_t^{(r)})\dot{\chi}_t^{(r)} + \partial_i h_\theta(\chi_t^{(r)}). \quad (9.11)$$

The observation mean, residual, and centered residual derivatives are

$$\dot{z}_t = \sum_{r=1}^M w_r \dot{z}_t^{(r)}, \quad (9.12)$$

$$\dot{v}_t = -\dot{\tilde{z}}_t, \quad (9.13)$$

$$\dot{\Delta}_t^{(r)} = \dot{z}_t^{(r)} - \dot{\tilde{z}}_t. \quad (9.14)$$

Differentiate S_t and $C_{xz,t}$:

$$\dot{S}_t = \dot{R}_\theta + \sum_{r=1}^M w_r \left\{ \dot{\Delta}_t^{(r)} \Delta_t^{(r)\top} + \Delta_t^{(r)} \dot{\Delta}_t^{(r)\top} \right\}, \quad (9.15)$$

$$\dot{C}_{xz,t} = \sum_{r=1}^M w_r \left\{ (\dot{\chi}_t^{(r)} - \dot{m}_t^-) \Delta_t^{(r)\top} + (\chi_t^{(r)} - m_t^-) \dot{\Delta}_t^{(r)\top} \right\}. \quad (9.16)$$

9.4 Innovation Score

Proposition 2 (Fixed Gaussian innovation score). *Assume S_t is positive definite and the active branch is differentiable. Let u_t solve*

$$S_t u_t = v_t. \quad (9.17)$$

Then the derivative of (7.13) is

$$\dot{\ell}_t^{\text{FSGQ}} = -\frac{1}{2} \left[\text{tr}(S_t^{-1} \dot{S}_t) + 2\dot{v}_t^\top u_t - u_t^\top \dot{S}_t u_t \right]. \quad (9.18)$$

Proof. Use

$$d \log \det S = \text{tr}(S^{-1} dS) \quad (9.19)$$

and

$$dS^{-1} = -S^{-1}(dS)S^{-1}. \quad (9.20)$$

Therefore, with $u = S^{-1}v$,

$$d(v^\top S^{-1}v) = 2(dv)^\top u - u^\top (dS)u. \quad (9.21)$$

Substitute $dS = \dot{S}_t$ and $dv = \dot{v}_t$ into (1.3). $\square$

9.5 Posterior Sensitivity Propagation

The gradient must also propagate $\dot{m}_t$ and $\dot{P}_t$ into the next time step. Differentiate $K_t = C_{xz,t} S_t^{-1}$:

$$\dot{K}_t = \dot{C}_{xz,t} S_t^{-1} - K_t \dot{S}_t S_t^{-1}. \quad (9.22)$$

Then

$$\dot{m}_t = \dot{m}_t^- + \dot{K}_t v_t + K_t \dot{v}_t, \quad (9.23)$$

$$\dot{P}_t = \dot{P}_t^- - \dot{K}_t S_t K_t^\top - K_t \dot{S}_t K_t^\top - K_t S_t \dot{K}_t^\top. \quad (9.24)$$

Equations (9.5)–(9.24) define the value-and-gradient recursion for the fixed scalar (8.2).

10 One Boxed Mathematical Algorithm

FixedSGQF value and gradient for one saved branch.

1. Choose dimension $b = n_x$, level L , one-dimensional rules I_ℓ , and merge tolerance. Build and save the cloud $\mathcal{C} = \{(\xi^{(r)}, w_r)\}_{r=1}^M$ by (5.1)–(5.5).
2. Save $y_{1:T}$, observation preprocessing, $m_0(\theta)$, $P_0(\theta)$, their sensitivity policy, the factor map $C(P)$, the symmetrize-then-veto rule, and model derivative routines.
3. Initialize m_0, P_0 and $\dot{m}_0, \dot{P}_0$ for each parameter component.
4. For $t = 1, \dots, T$, run the prediction value equations (7.2)–(7.5) and prediction sensitivity equations (9.5)–(9.9).
5. If P_t^- violates the saved branch or veto policy, stop and report the branch failure.
6. Run the observation value equations (7.6)–(7.13) and observation sensitivity equations (9.10)–(9.15).
7. If S_t violates the saved positive-definite branch, stop and report the branch failure.
8. Add the score (9.18) to the running gradient and add $\hat{\ell}_t^{\text{FSGQ}}$ to the running scalar.
9. Update $m_t, P_t, \dot{m}_t, \dot{P}_t$ using (7.14)–(7.16) and (9.22)–(9.24).

11 Implementation Contract

object	definition	shape	reuse
$\xi^{(r)}$	standardized sparse-grid node	n_x	value and gradient
w_r	accumulated signed weight after duplicate merge	scalar	value and gradient
C_{t-1}, C_t^-	saved covariance factors on declared branch	$n_x \times n_x$	value, gradient, diagnostics
$\chi_{t-1}^{(r)}$	previous placed point	n_x	transition value and derivative
$a_t^{(r)}$	transition image	n_x	prediction moments
$\chi_t^{(r)}$	predictive placed point	n_x	observation value and derivative
$z_t^{(r)}$	observation image	n_y	observation moments
$\bar{z}_t, S_t, C_{xz,t}$	Gaussian projection moments	$n_y, n_y \times n_y, n_x \times n_y$	likelihood and update
v_t, K_t	innovation and gain	$n_y, n_x \times n_y$	likelihood, update, gradient
$\dot{\chi}, \dot{a}, \dot{z}$	point sensitivities	matching objects	value gradient only
$\dot{m}, \dot{P}, \dot{S}, \dot{C}_{xz}$	moment and filter sensitivities	matching objects	value gradient and next step
data record	$y_{1:T}$, preprocessing, missing-data policy if any	metadata	value, gradient, finite difference
initial law	$m_0(\theta), P_0(\theta), \dot{m}_0, \dot{P}_0$	$n_x, n_x \times n_x$	value and gradient start
branch record	cloud, merge tolerance, factor branch, symmetrize-then-veto rule	metadata	same-scalar finite difference

12 Finite-Difference Same-Scalar Check

The finite-difference check must call the same value routine with the same saved branch $\mathcal{B}$. For parameter component i , use

$$D_i(h) = \frac{\widehat{\ell}_T^{\text{FSGQ}}(\theta + he_i; \mathcal{B}) - \widehat{\ell}_T^{\text{FSGQ}}(\theta - he_i; \mathcal{B})}{2h}. \quad (12.1)$$

Use the step ladder

$$h \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\} \max\{1, |\theta_i|\}. \quad (12.2)$$

For every h , the $+h$ and $-h$ evaluations must reuse the same cloud, preprocessing, initial-condition policy, factor branch, merge tolerance, and symmetrize-then-veto rule. If either side hits a different branch or returns a veto, that row is marked branch-invalid and is not interpreted as a derivative check. For valid rows, compare $D_i(h)$ against the analytical score G_i by

$$e_i(h) = \frac{|D_i(h) - G_i|}{\max\{1, |D_i(h)|, |G_i|\}}. \quad (12.3)$$

The diagnostic passes when at least two consecutive valid rows have $e_i(h) \leq 10^{-5}$ and the smaller-step row is no worse than ten times the larger-step row. This is a numerical diagnostic, not a theorem: failure means one of three things is likely. The derivative formula is wrong, the value and gradient paths are not using the same scalar, or the branch is too nonsmooth or ill-conditioned at that point.

A scalar numerical trace. Let $X \sim \mathcal{N}(0, 1)$, $h_\theta(x) = \theta x$, $R = 1$, and $y = 2$. Any fixed Gaussian rule with exact second moment gives

$$\bar{z} = 0, \quad S(\theta) = 1 + \theta^2, \quad \hat{\ell}(\theta) = -\frac{1}{2} \left\{ \log(1 + \theta^2) + \frac{4}{1 + \theta^2} + \log(2\pi) \right\}. \quad (12.4)$$

At $\theta = 1$,

$$G = -\frac{1}{2} \left\{ \frac{2}{2} - \frac{4 \cdot 2}{2^2} \right\} = 0.5. \quad (12.5)$$

Central differences give:

h	$D(h)$	G	$ D(h) - G $
10^{-1}	0.5008108250	0.5	8.10825×10^{-4}
10^{-2}	0.5000083311	0.5	8.33108×10^{-6}
10^{-3}	0.5000000833	0.5	8.33331×10^{-8}
10^{-4}	0.5000000008	0.5	8.33722×10^{-10}

A cloud-sensitive nonlinear trace. The previous trace tests the innovation-score algebra. To exercise the fixed cloud, use the two-dimensional level-2 cloud in Section 6 and the nonlinear observation

$$h_\theta(x_1, x_2) = \theta(x_1^2 + x_2^2), \quad X \sim \mathcal{N}(0, I_2), \quad R = 1, \quad y = 3. \quad (12.6)$$

For the merged cloud in Section 6,

$$\sum_r w_r(x_{1,r}^2 + x_{2,r}^2) = 2, \quad \sum_r w_r(x_{1,r}^2 + x_{2,r}^2)^2 = 6. \quad (12.7)$$

Therefore

$$\bar{z}(\theta) = 2\theta, \quad S(\theta) = 1 + 2\theta^2, \quad v(\theta) = 3 - 2\theta, \quad (12.8)$$

if the parameter scales the whole nonlinear observation linearly. To exercise the parameter through placed nonlinear values more sharply, set instead

$$h_\theta(x_1, x_2) = \theta^2(x_1^2 + x_2^2), \quad (12.9)$$

which gives

$$\bar{z}(\theta) = 2\theta^2, \quad S(\theta) = 1 + 2\theta^4, \quad v(\theta) = 3 - 2\theta^2. \quad (12.10)$$

At $\theta = 1$, the score from (9.18) is

$$G = -\frac{1}{2} \left[\frac{8}{3} + 2(-4)\frac{1}{3} - \frac{1}{3^2}8 \right] = \frac{4}{9}. \quad (12.11)$$

Central differences for the saved cloud scalar give:

h	$D(h)$	G	$ D(h) - G $
10^{-1}	0.5200303321	0.4444444444	7.55859×10^{-2}
10^{-2}	0.4452146703	0.4444444444	7.70226×10^{-4}
10^{-3}	0.4444521481	0.4444444444	7.70369×10^{-6}
10^{-4}	0.4444445215	0.4444444444	7.70368×10^{-8}

This example depends on the merged sparse-grid fourth moment in (12.7). A duplicate-merge bug or a different saved cloud changes $S(\theta)$, G , and the finite-difference table.

13 Diagnostics, Accuracy, Memory, And Performance Tests

Fixed SGQF should be tested as an approximate deterministic likelihood, not as an exact posterior method. The minimum validation suite is:

1. **Construction checks.** Verify duplicate-node accumulation, weight totals, signed-weight counts, and deterministic cloud reproduction for small (b, L) cases such as Section 6.
2. **Linear-Gaussian recovery.** For affine f_θ, h_θ , the Gaussian projection recursion should match the Kalman filter up to quadrature and floating-point tolerance.
3. **Quadratic scalar cell.** Use (1.5) to show both the moment accuracy and the posterior-shape limitation.
4. **Same-scalar finite differences.** Run (12.1) for representative parameters and step sizes.
5. **Signed-weight stability.** Record the minimum eigenvalues of symmetrized P_t^- , S_t , and P_t , plus any branch veto.
6. **Memory and runtime scaling.** Record $M_{b,L}$, model evaluations, wall time, and peak memory as b, L, T change.
7. **Accuracy ladder.** Compare fixed SGQF against tensor-product GHQ in small dimension, Kalman truth in linear-Gaussian models, and particle or TT diagnostics in nonlinear models where those references are available.

Passing these checks supports the statement that the implementation evaluates and differentiates the declared approximate scalar. It does not prove exact posterior accuracy.

14 Adaptive Sparse Grids As Grid Design

Singh et al. [2] choose sparse-grid indices adaptively by using admissible index sets, local error indicators, active and old index sets, a global error estimate, and a tolerance. That is useful for designing a cloud before inference. For HMC, however, the live target cannot silently change its grid as θ moves. The safe interpretation is

$$\text{adaptive pilot chooses a cloud} \implies \text{freeze the cloud} \implies \text{run FixedSGQF}. \quad (14.1)$$

If the adaptive index set is rerun inside the likelihood, the value path is a piecewise-defined algorithm with discrete changes. The formulas in Section 9 do not differentiate those changes.

15 Relation To Zhao–Cui Squared TT

Fixed SGQF and Zhao–Cui squared TT solve different parts of the high-dimensional filtering problem.

question	Fixed SGQF	Zhao–Cui squared TT
stored object	one Gaussian mean/covariance and an approximate innovation scalar	nonnegative TT approximation of filtering/posterior densities
main strength	deterministic fixed cloud, transparent scalar, analytical same-scalar gradient	richer non-Gaussian density representation and conditional maps
main limitation	Gaussian projection cannot store multimodality or skewed posterior shape	adaptive TT-cross/rank changes are not globally smooth; implementation is heavier
gradient lane	differentiates fixed Gaussian-projection likelihood	fixed-branch derivative differentiates a declared TT approximate scalar
best use	fast approximate likelihood, baseline, local diagnostic, candidate fixed target when branch diagnostics stay stable	high-dimensional non-Gaussian filtering and smoothing proposal

The proposal is therefore not to replace Zhao–Cui with fixed SGQF. The proposal is to carry two honest candidates: a transparent fixed-rule Gaussian projection likelihood with a same-scalar gradient, and a richer squared-TT density approximation for the cases where Gaussian projection is too weak.

16 Where Each Construction Comes From

construction	support	use in this note
state-space and conceptual filtering equations	Jia et al. [1, Section 2]	Sections 2–3
Gaussian approximation filter moment structure	Jia et al. [1, Eq. 15–23] plus Proposition 1	moment projection and update equations
one-dimensional and tensor-product Gaussian quadrature	Jia et al. [1, Section 2.2.1]	Section 4
sparse-grid formula and index band	Jia et al. [1, Eq. 26–29]	Section 5
point/weight generation and duplicate accumulation	Jia et al. [1, Algorithm 1]	fixed cloud $\mathcal{C}$ and toy grid
quadrature exactness and UKF relation	Jia et al. [1, Theorems 3.1–3.2]	source-scope orientation, not posterior guarantee
adaptive sparse-grid mechanics	Singh et al. [2, Section 3]	offline grid-design interpretation
fixed scalar and analytical gradient comparison with squared TT	BayesFilter project derivation Zhao–Cui companion note and Zhao and Cui [3]	Sections 8–9 Section 15

17 Conclusion

FixedSGQF is now a well-defined standalone proposal: choose and save a sparse-grid cloud, evaluate the Gaussian projection likelihood, differentiate that same saved scalar, and reject branch changes that would make the value and gradient disagree. Its intellectual virtue is clarity. Its mathematical limitation is also clear: it is still a Gaussian projection filter. That honesty is what makes it a useful companion to the Zhao–Cui squared-TT lane rather than a weaker substitute for it.

References

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- [2] Abhinoy Kumar Singh, Rahul Radhakrishnan, Shovan Bhaumik, and Paresh Date. Adaptive sparse-grid gauss-hermite filter. *arXiv preprint arXiv:1803.09272*, 2018.
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